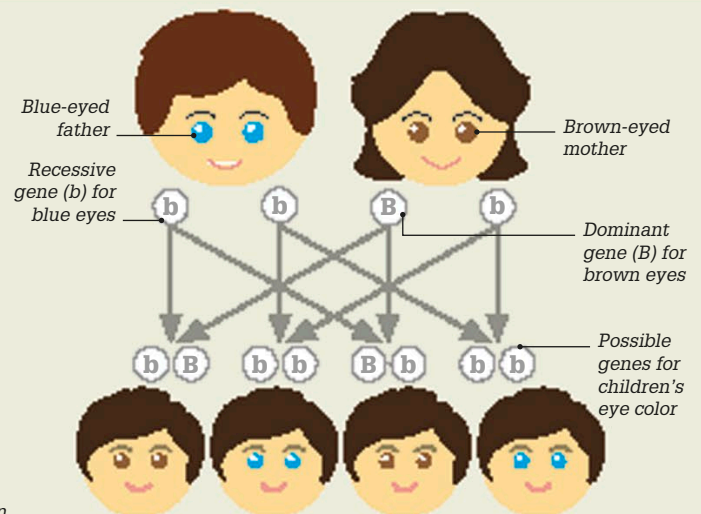


The code of life

Parents pass on traits from themselves to their offspring—from the shape of their nose to the likelihood of them suffering from certain diseases. The instructions for these traits and how the offspring should develop are called genes, which are stored in a chemical called DNA (deoxyribonucleic acid), present in every cell. Genetics is the study of how genes work and are passed down from one generation to another.



Passing things down

Everyone has two versions of each gene, one from their mother and one from their father. Some of these genes, such as brown eye color (B) can be dominant, meaning that if you have one brown eye gene and one blue eye gene (b), the dominant gene wins out and you have brown eyes.

DNA

Genes are carried inside cells in DNA. Long ribbons of DNA are formed from two spiraling chains, together called a double helix. Linking the chains together are strands containing pairs of four chemicals, called bases—guanine, cytosine, thymine, and adenine. These base pairs form a four-letter alphabet that acts as a code, telling cells how to make proteins.

DNA's "backbone" is made of carbon, hydrogen, oxygen, and phosphorus.

Key events

1866

Austrian botanist Gregor Mendel's work on pea plants enabled his discovery of the key laws of inheritance, showing how certain traits are passed on by plants to their offspring.

1911

Through his studies on the chromosomes of fruit flies, American biologist Thomas Hunt Morgan demonstrated that chromosomes carry the genes of a species.



1951

British chemist Rosalind Franklin photographed DNA fibers for the first time in X-ray studies involving her colleagues Maurice Wilkins and Raymond Gosling.

1953

American geneticist James Watson and British biologist Francis Crick published evidence of the double helix structure of the DNA.